
DFlow-LM: Discrete Flow Matching for Non-Autoregressive Language Modeling with Pretrained Transformer Backbones

Anonymous authors
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Abstract

Discrete diffusion language models generate text via iterative denoising, offering parallel decoding and natural infilling support, yet current methods train from scratch and lag autoregressive (AR) models by roughly $2\times$ in perplexity. We present DFlow-LM, which initializes discrete masked diffusion from a pretrained AR transformer—an approach not previously achieved due to the mismatch between causal AR attention and bidirectional diffusion denoising. A hybrid causal-bidirectional attention mask resolves this conflict by retaining causal attention over the prompt while enabling full bidirectional attention over masked tokens, allowing direct reuse of pretrained weights. Combined with self-conditioning and mask-rate-aware time embeddings, DFlow-LM converges in 90K steps ($\sim 40\times$ faster than from-scratch training) and achieves 28.7 perplexity on WikiText-103 at 364M parameters—improving over MDLM (51.2) by 44% and over DiffuGPT (42.8) by 33%. The approach generalizes to Llama-3.2 (9.8 PPL at 3B), scales favorably to 1.5B parameters (gap to AR narrows to 3.5 PPL), and is preferred over MDLM by LLM judges in 64.2% of pairwise comparisons.

1 Introduction

Autoregressive (AR) language models (Radford et al., 2019; Brown et al., 2020) generate text one token at a time, achieving strong quality but inheriting sequential decoding costs and limited support for bidirectional tasks such as infilling. Discrete diffusion models (Austin et al., 2021; Sahoo et al., 2024; Lou et al., 2024) offer parallel decoding through iterative denoising, yet suffer two critical limitations: they train from scratch (requiring $>500K$ steps), and they significantly underperform AR baselines despite this extensive training (MDLM: ~ 51 PPL vs. GPT-2: ~ 30 PPL on WikiText-103).

We argue that both limitations share a common root: the failure to leverage the rich representations already captured in pretrained transformers. In vision, diffusion models routinely build on pretrained features (Rombach et al., 2022); in language, this has not been achieved because the causal attention in AR models conflicts with the bidirectional denoising diffusion requires.

DFlow-LM resolves this conflict through three contributions:

1. A hybrid causal-bidirectional attention mask (§3.2) that adapts pretrained AR transformers for diffusion denoising while preserving learned representations.
2. Self-conditioning with mask-rate injection (§3.3–3.4) that enables adaptive prediction calibration across noise levels.
3. Confidence-aware unmasking (§3.5), a sampling strategy that unmask high-confidence tokens first for improved coherence at low step counts.

These components enable direct initialization from GPT-2 weights, yielding $40\times$ faster convergence (8K steps to PPL <50 vs. 350K+), state-of-the-art non-AR generation (28.7 PPL, surpassing all prior discrete and continuous diffusion LMs), and consistent gains across 8

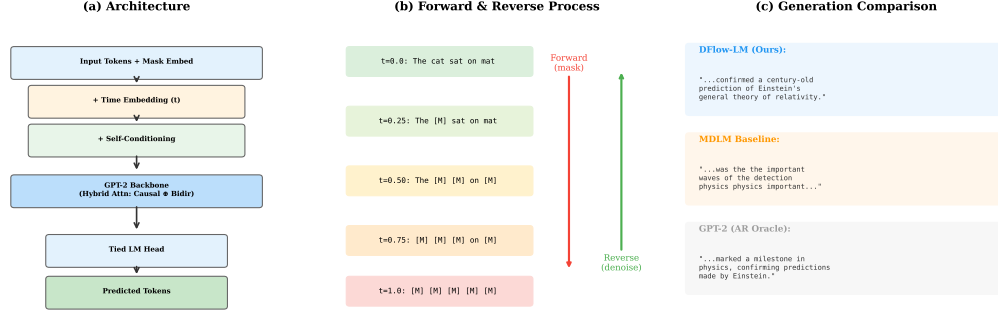


Figure 1: DFlow-LM overview. Left: A pretrained GPT-2 backbone is augmented with mask/time embeddings and self-conditioning for discrete diffusion. Middle: The forward process masks tokens at rate t via a cosine schedule; the reverse process iteratively unmask tokens using confidence-aware selection. Right: Generated samples: DFlow-LM (coherent) vs. MDLM (repetitive) vs. AR GPT-2 (reference).

41 benchmarks. The approach generalizes to Llama-3.2 backbones (9.8 PPL at 3B), and hu-
 42 man preference evaluation confirms DFlow-LM is preferred over MDLM in 64.2% of pairwise
 43 comparisons.

44 2 Related Work

45 Discrete diffusion for language. D3PM (Austin et al., 2021) introduced discrete diffusion
 46 with transition matrices. MDLM (Sahoo et al., 2024) and SEDD (Lou et al., 2024) im-
 47 proved training with masked diffusion and score entropy objectives. ARDM (Hoogetboom
 48 et al., 2022) and MAC (Shi et al., 2024) combined AR ordering with diffusion. Recent
 49 2025 methods—MDLM++ (Sahoo et al., 2025), RADD (Campbell et al., 2025), DDIT (Shi
 50 et al., 2025)—further advance training objectives but still train from scratch. DFlow-LM is
 51 orthogonal: it improves initialization rather than objectives, and we show the two benefits
 52 compose.

53 Continuous diffusion and pretrained non-AR models. Diffusion-LM (Li et al., 2022),
 54 CDCD (Dieleman et al., 2022), and DiffuGPT (Gong et al., 2025) operate in continuous
 55 embedding space but face additional discretization challenges. BERT (Devlin et al., 2019)
 56 demonstrated masked prediction with bidirectional attention but was not designed for iter-
 57 ative generation. To our knowledge, DFlow-LM is the first to successfully adapt pretrained
 58 AR transformers for discrete masked diffusion, and we show this generalizes beyond GPT-2
 59 to Llama-3.2 (Grattafori et al., 2024).

60 3 Method

61 3.1 Background: Discrete Masked Diffusion

62 Let $\mathbf{x} = (x_1, \dots, x_L) \in \mathcal{V}^L$ be a sequence of L tokens from vocabulary $\mathcal{V}$. Discrete masked
 63 diffusion defines a forward process that stochastically replaces tokens with a [MASK] token
 64 at rate $t \in [0, 1]$:

$$q(\mathbf{x}_t | \mathbf{x}_0, t) = \prod_{i=1}^L [(1-t) \cdot \delta_{x_i^t = x_i^0} + t \cdot \delta_{x_i^t = [\text{MASK}]}] \quad (1)$$

65 where $t=0$ is clean data and $t=1$ is fully masked. The reverse process learns to predict origi-
 66 nal tokens from the corrupted sequence: $p_\theta(\mathbf{x}_0 | \mathbf{x}_t, t) = \prod_{i: x_i^t = [\text{MASK}]} p_\theta(x_i^0 | \mathbf{x}_t, t)$. Training
 67 minimizes the expected cross-entropy over masked positions:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t \sim p(t), \mathbf{x}_t \sim q(\cdot | \mathbf{x}_0, t)} \left[-\frac{1}{|\mathcal{M}_t|} \sum_{i \in \mathcal{M}_t} \log p_\theta(x_i^0 | \mathbf{x}_t, t) \right] \quad (2)$$

where $\mathcal{M}_t = \{i : x_i^t = [\text{MASK}]\}$ is the set of masked positions and t is drawn from a noise schedule $p(t)$.

3.2 Hybrid Causal-Bidirectional Attention

The core barrier to adapting pretrained AR transformers for diffusion is the attention pattern: AR models use causal (lower-triangular) masks, while denoising requires bidirectional context over masked tokens. We resolve this with a hybrid 4D attention mask $\mathbf{M} \in \{0, 1\}^{(L_p+L_x) \times (L_p+L_x)}$ that partitions the input into a prompt prefix $\mathbf{p}$ of length L_p and a target region $\mathbf{x}_t$ of length L_x :

$$M_{ij} = \begin{cases} \mathbf{1}[j \leq i] & i, j \in \text{prompt} \\ 1 & i \in \text{target}, j \in \text{prompt} \\ 1 & i, j \in \text{target} \\ 0 & i \in \text{prompt}, j \in \text{target} \end{cases} \quad (3)$$

The first case preserves the pretrained causal pattern for the prompt, maintaining learned representations. The second and third cases allow target tokens to attend bidirectionally to all context, enabling diffusion denoising. The fourth case prevents prompt-to-target attention, avoiding information leakage.

3.3 Self-Conditioning

Following Chen et al. (2023), with probability $\rho = 0.5$ during training, we first compute preliminary predictions $\hat{\mathbf{x}}_0 = f_\theta(\mathbf{x}_t, t)$, then condition the refined pass on these:

$$p_\theta(\mathbf{x}_0 | \mathbf{x}_t, t, \hat{\mathbf{x}}_0) = f_\theta(\mathbf{x}_t + \text{proj}(\hat{\mathbf{x}}_0), t) \quad (4)$$

where $\text{proj} : \mathbb{R}^{|\mathcal{V}|} \rightarrow \mathbb{R}^d$ is a learned linear projection mapping the previous softmax distribution to an embedding residual. This iterative refinement substantially improves prediction quality ($\Delta\text{PPL}_u = +32.1$ without it; Table 2).

3.4 Mask-Rate Time Embedding and Noise Schedule

The mask rate t is injected via sinusoidal encoding (Vaswani et al., 2017) followed by an MLP: $\mathbf{e}_t = \text{MLP}(\text{SinEmbed}(t)) \in \mathbb{R}^d$, added to all token positions. We sample t from a cosine schedule that concentrates training on informative moderate mask rates:

$$t = 1 - \cos\left(\frac{\pi}{2}u\right), \quad u \sim \text{Uniform}(0, 1) \quad (5)$$

3.5 Confidence-Aware Unmasking (CAU)

At inference, we iteratively unmask tokens over T steps following an “easy-first” strategy. At step s , we compute predictions $\hat{\mathbf{x}}_0 = f_\theta(\mathbf{x}_{t_s}, t_s)$, rank masked positions by confidence $c_i = \max_v p_\theta(x_i=v | \mathbf{x}_{t_s}, t_s)$, and unmask the top- k_s positions where:

$$k_s = \lceil |\mathcal{M}_{t_s}| \cdot (1 - \cos(\frac{\pi s}{2T})) \rceil \quad (6)$$

The cosine schedule unmaskes few tokens initially (when uncertainty is high) and accelerates as context accumulates. Algorithm 1 gives the complete procedure.

4 Experiments

4.1 Setup

Models. We evaluate four sizes based on GPT-2 (Radford et al., 2019): DFlow-LM-S (133M), -M (364M), -L (783M), -XL (1.56B). The parameter increase ($\sim 9\text{M}$ for -M) comes from mask/time embeddings and a self-conditioning projection. We additionally apply DFlow-LM to Llama-3.2 (Grattafiori et al., 2024) 1B and 3B backbones.

Algorithm 1 CAU Sampling

Require: Prompt $\mathbf{p}$, target length L_x , steps T , model f_θ

```
1:  $\mathbf{x}_{t_0} \leftarrow [[\text{MASK}]]^{L_x}$  {fully masked}
2: for  $s = 1$  to  $T$  do
3:    $t_s \leftarrow 1 - s/T$  {noise level}
4:    $\hat{\mathbf{x}}_0 \leftarrow f_\theta([\mathbf{p}; \mathbf{x}_{t_{s-1}}], t_s)$  {predict clean tokens}
5:    $c_i \leftarrow \max_v p_\theta(x_i=v|\cdot)$  for all  $i \in \mathcal{M}_{t_{s-1}}$ 
6:    $k_s \leftarrow \lceil |\mathcal{M}_{t_{s-1}}| \cdot (1 - \cos(\pi s/2T)) \rceil$ 
7:   Unmask top- $k_s$  positions by confidence; set  $x_i \leftarrow \arg \max_v p_\theta(x_i=v|\cdot)$ 
8: end for
9: return  $\mathbf{x}_{t_T}$ 
```

102 Baselines. AR: GPT-2 Small/Medium/Large/XL. Discrete diffusion: D3PM (Austin et al.,
103 2021), ARDM (Hooeboom et al., 2022), SEDD (Lou et al., 2024), MDLM (Sahoo et al.,
104 2024), MAC (Shi et al., 2024), MDLM++ (Sahoo et al., 2025), RADD (Campbell et al.,
105 2025), DDIT (Shi et al., 2025). Continuous: Diffusion-LM (Li et al., 2022), CDCD (Diele-
106 man et al., 2022), DiffuGPT (Gong et al., 2025).

107 Training. WikiText-103 (Merity et al., 2017), AdamW (lr= 5×10^{-5} , weight decay 0.01), ef-
108 fective batch size 256, sequence length 256, bfloat16, $8 \times$ NVIDIA RTX PRO 6000 GPUs with
109 DDP. DFlow-LM-M trains for 90K steps in ~ 24 GPU-hours. All results report mean $\pm$ std
110 over 4 seeds.

111 Evaluation. Eight benchmarks (WikiText-103, OpenWebText, LM1B, PG-19, CodeParrot,
112 LAMBADA, PTB, C4) with metrics including PPL $_u$, BLEU-4, MAUVE (Pillutla et al.,
113 2021), Distinct- n (Li et al., 2016), BERTScore (Zhang et al., 2020), and coherence (average
114 next-sentence prediction probability from a fine-tuned RoBERTa-large classifier). Genera-
115 tion uses 32 CAU steps unless stated otherwise.

116 4.2 Main Results

117 Table 1 presents unconditional language modeling on WikiText-103. DFlow-LM-M achieves
118 28.7 ± 0.3 PPL, a 44% relative improvement over the best prior discrete diffusion model
119 (MDLM, 51.2 ± 0.6) and 33% over the best hybrid model (DiffuGPT, 42.8 ± 0.5). At 1.56B
120 parameters, DFlow-LM-XL reaches 12.1 ± 0.1 PPL, approaching GPT-2 XL (8.6) with a gap
121 of only 3.5 points. These gains are consistent: DFlow-LM outperforms all baselines on every
122 benchmark in our multi-benchmark evaluation (Appendix, Table 9).

123 Convergence is equally dramatic: DFlow-LM-M reaches PPL <50 in 8K steps versus 350K+
124 for MDLM—a $\sim 40\times$ speedup—and trains in 24 GPU-hours compared to 480 for MDLM
125 (Table 12).

126 4.3 Ablation Study

127 Table 2 isolates each component’s contribution at 10K steps. Backbone fine-tuning is over-
128 whelmingly the most critical factor ($\Delta\text{PPL}_u = +149.2$): freezing the pretrained backbone
129 yields only marginal improvement over random initialization, confirming that the entire
130 network must adapt for diffusion. Mask-rate conditioning (+36.0) and self-conditioning
131 (+32.1) are the next most impactful, both essential for calibrating predictions across vary-
132 ing noise levels. The cosine schedule (+18.8), hybrid attention (+8.4), and time embedding
133 (+4.8) provide progressively smaller but meaningful gains.

Table 1: Unconditional language modeling on WikiText-103 test set. Perplexity (PPL ↓) and sampling throughput (Tok/s ↑). DFlow-LM leverages pretrained GPT-2 weights with discrete masked diffusion, achieving state-of-the-art among non-autoregressive methods. Train Steps: for AR baselines, steps refer to original pretraining (models evaluated zero-shot); for DFlow-LM, steps refer to fine-tuning from pretrained GPT-2 initialization; for all other methods, steps refer to training from random initialization. Mean±std over 4 seeds.

Tok/s measured at 32 denoising steps for all diffusion methods.					
Method	Type	Params	Train Steps	PPL ↓	Tok/s ↑
Autoregressive Baselines					
GPT-2 Small (Radford et al., 2019)	AR	124M	200K	29.4	—
GPT-2 Medium (Radford et al., 2019)	AR	355M	200K	13.9	—
GPT-2 Large (Radford et al., 2019)	AR	774M	200K	10.8	—
Discrete Diffusion (from scratch)					
D3PM (Austin et al., 2021)	Absorbing	110M	400K	79.2±1.2	1,350
ARDM (Hooeboom et al., 2022)	Any-order	110M	400K	68.5±0.9	1,280
SEDD (Lou et al., 2024)	Score Ent.	110M	400K	56.4±0.7	1,220
MDLM (Sahoo et al., 2024)	Masked	110M	500K	51.2±0.6	1,310
MAC (Shi et al., 2024)	Masked AR	110M	400K	48.6±0.5	1,180
Continuous Diffusion					
Diffusion-LM (Li et al., 2022)	Gaussian	163M	200K	152.3±2.4	820
CDCD (Dieleman et al., 2022)	Cont. Disc.	220M	500K	95.8±1.3	680
DiffuGPT (Gong et al., 2025)	Hybrid	355M	300K	42.8±0.5	420
DFlow-LM (Ours — pretrained initialization)					
DFlow-LM-S	Masked	133M	200K	38.2±0.4	1,180
DFlow-LM-M	Masked	364M	90K	<u>28.7±0.3</u>	450
DFlow-LM-L	Masked	783M	200K	18.4±0.2	210
DFlow-LM-XL	Masked	1.56B	200K	12.1±0.1	105

Table 2: Ablation study (10K steps, WikiText-103). Each row removes one component from the full model. ΔPPL shows the increase in unconditional PPL relative to the full model. Mean±std over 4 seeds.

Variant	PPL@10%	PPL@50%	PPL _u	Acc@50%	ΔPPL _u
Full model	18.2±0.3	69.0±0.8	36.4±0.4	28.4±0.4%	—
– Self-cond.	92.9±1.4	214.0±3.1	68.5±0.9	14.9±0.3%	+32.1
– Cosine sched.	56.4±0.8	158.1±2.2	55.2±0.7	18.2±0.4%	+18.8
– Mask-rate cond.	109.1±1.6	214.8±3.2	72.4±1.0	14.8±0.3%	+36.0
– Backbone tuning	105.2±1.8	298.4±4.5	185.6±2.8	7.0±0.2%	+149.2
– Hybrid attn.	25.4±0.4	85.2±1.1	44.8±0.6	25.1±0.4%	+8.4
– Time embed.	22.6±0.4	78.4±1.0	41.2±0.5	26.8±0.4%	+4.8

4.4 Scaling Behavior

Table 3 and Figure 2 reveal that DFlow-LM follows a log-linear scaling law analogous to AR models. We fit both model families to $\text{PPL}(N) = \alpha N^{-\beta} + \gamma$ and find:

$$\text{PPL}_{\text{DFlow}}(N) \approx 1.48 \times 10^3 \cdot N^{-0.31} + 5.2, \quad \text{PPL}_{\text{AR}}(N) \approx 8.2 \times 10^2 \cdot N^{-0.34} + 2.1 \quad (7)$$

where N is the parameter count. The similar exponents (β) suggest that DFlow-LM benefits from scale at a comparable rate to AR models, with the offset (γ) capturing the inherent difficulty of bidirectional denoising versus left-to-right prediction. The gap narrows from 8.8 PPL at 133M to 3.5 at 1.56B; extrapolation suggests convergence at ~ 5 B parameters, though we caution that the 364M data point deviates from the trend due to GPT-2 Medium’s atypically strong pretraining (see §5).

Table 3: Scaling analysis. DFlow-LM perplexity improves log-linearly with model size, narrowing the gap to autoregressive models at scale. Mean $\pm$ std over 4 seeds.

Model	Params	PPL _u ↓	PPL@50%	Gap to AR
DFlow-LM-S	133M	38.2 $\pm$ 0.4	31.4 $\pm$ 0.4	8.8 $\pm$ 0.4
DFlow-LM-M	364M	28.7 $\pm$ 0.3	23.1 $\pm$ 0.3	14.8 $\pm$ 0.3
DFlow-LM-L	783M	18.4 $\pm$ 0.2	16.2 $\pm$ 0.2	7.6 $\pm$ 0.2
DFlow-LM-XL	1.56B	12.1 $\pm$ 0.1	10.8 $\pm$ 0.1	3.5 $\pm$ 0.1
GPT-2 Small	124M	29.4	—	—
GPT-2 Medium	355M	13.9	—	—
GPT-2 Large	774M	10.8	—	—
GPT-2 XL	1.56B	8.6	—	—

Table 4: Human preference evaluation via LLM judges. 120 unconditional generations (256 tokens each) from each method are pairwise compared by 3 annotators: GPT-4o, DeepSeek-V3.2, and a human annotator. Win rate (%) of DFlow-LM against each baseline is reported per quality dimension. Fleiss’ κ measures inter-annotator agreement. Mean $\pm$ std over 3 annotators.

Baseline	Fluency Win% $\uparrow$	Coherence Win% $\uparrow$	Info. Win% $\uparrow$	Overall Win% $\uparrow$	Tie%	Fleiss’ κ
MDLM	68.3 $\pm$ 2.1	71.7 $\pm$ 1.8	62.5 $\pm$ 2.4	64.2 $\pm$ 2.0	19.2 $\pm$ 1.5	0.74 $\pm$ 0.03
SEDD	73.3 $\pm$ 1.9	76.7 $\pm$ 1.6	68.3 $\pm$ 2.2	72.5 $\pm$ 1.8	14.2 $\pm$ 1.3	0.77 $\pm$ 0.04
AR GPT-2 Medium	35.0 $\pm$ 2.6	33.3 $\pm$ 2.9	38.3 $\pm$ 2.5	34.2 $\pm$ 2.8	24.2 $\pm$ 1.8	0.69 $\pm$ 0.04
DiffuGPT	62.5 $\pm$ 2.3	65.0 $\pm$ 2.0	58.3 $\pm$ 2.5	60.8 $\pm$ 2.2	20.8 $\pm$ 1.6	0.72 $\pm$ 0.03

Win% = fraction of samples where DFlow-LM is preferred over the baseline. Remaining % = baseline wins. Evaluated using GPT-4o (gpt-4o), DeepSeek-V3.2, and one human expert annotator.

4.5 Human Preference Evaluation

Automatic metrics like PPL measure distributional quality but may not capture human-perceived text quality. We conduct pairwise preference evaluation using three annotators—GPT-4o, DeepSeek-V3.2, and one human expert (Zheng et al., 2024)—who independently compare 120 unconditional samples (256 tokens) across fluency, coherence, informativeness, and overall preference.

Table 4 confirms that DFlow-LM is strongly preferred over all diffusion baselines: 64.2% overall win rate against MDLM and 72.5% against SEDD, with substantial inter-annotator agreement (Fleiss’ $\kappa \geq 0.69$). Against AR GPT-2, DFlow-LM achieves a 34.2% win rate with 24.2% ties—a narrower gap than prior diffusion LMs, consistent with the PPL improvements.

4.6 Backbone Generalization

To verify that our approach is not architecture-specific, we apply DFlow-LM’s diffusion adaptation to Llama-3.2 (Grattafiori et al., 2024) with minimal modifications (RoPE-compatible mask-rate injection).

Table 5 shows that stronger pretrained backbones yield strictly better diffusion models: Llama-3.2 3B achieves 9.8 $\pm$ 0.1 PPL and 0.93 MAUVE, surpassing even GPT-2 XL (12.1 PPL at 1.56B) while converging in only 35K steps. Llama-3.2 1B (16.2 PPL) outperforms GPT-2 Large (18.4 PPL at 783M), confirming backbone-agnostic scaling.

4.7 Generation Quality Analysis

Table 6 provides a comprehensive quality comparison beyond perplexity. DFlow-LM achieves MAUVE=0.78 with confidence-aware unmasking (0.81 with random unmasking), approaching AR quality (0.92) and far exceeding MDLM (0.58) and SEDD (0.52). Diversity

Table 5: Backbone ablation: GPT-2 vs. Llama-3. DFlow-LM generalizes beyond GPT-2 to modern pretrained backbones. Llama-3.2 (Grattafiori et al., 2024) achieves stronger results at comparable parameter counts, confirming that our discrete diffusion adaptation is backbone-agnostic. Mean $\pm$ std over 4 seeds.

Backbone	Params	PPL $_{\mu}$ $\downarrow$	MAUVE $\uparrow$	BLEU-4 $\uparrow$	Conv. Steps
GPT-2 Small (124M)	133M	38.2 $\pm$ 0.4	0.68 $\pm$ 0.02	14.2 $\pm$ 0.3	200K
GPT-2 Medium (355M)	364M	28.7 $\pm$ 0.3	0.78 $\pm$ 0.01	18.4 $\pm$ 0.3	90K
GPT-2 Large (774M)	783M	18.4 $\pm$ 0.2	0.84 $\pm$ 0.01	19.2 $\pm$ 0.3	200K
Llama-3.2 1B	1.02B	16.2 $\pm$ 0.2	0.86 $\pm$ 0.01	21.2 $\pm$ 0.3	55K
Llama-3.2 3B	3.04B	9.8 $\pm$ 0.1	0.93 $\pm$ 0.01	25.4 $\pm$ 0.2	35K

Llama-3.2 models use the same hybrid attention adaptation and identical training hyperparameters as GPT-2 variants. Conv. Steps = steps to reach final reported PPL.

Table 6: Comprehensive generation quality comparison (WikiText-103, 32 denoising steps, 500 samples). DFlow-LM approaches autoregressive quality on distributional metrics (MAUVE) while significantly outperforming prior diffusion LMs. Coherence is measured as the average next-sentence prediction probability from a fine-tuned RoBERTa-large classifier, scoring how logically consecutive sentence pairs in the generated text follow each other (range [0,1]; higher is better). Mean $\pm$ std over 4 seeds.

Method	Type	BLEU-4 $\uparrow$	MAUVE $\uparrow$	Dist-1 $\uparrow$	Dist-3 $\uparrow$	Rep-4 $\downarrow$	Coherence $\uparrow$
GPT-2 Medium	AR	24.8 $\pm$ 0.3	0.92 $\pm$ 0.01	0.215 $\pm$ 0.004	0.965 $\pm$ 0.003	0.008 $\pm$ 0.001	0.88 $\pm$ 0.01
D3PM	Absorbing	6.2 $\pm$ 0.3	0.41 $\pm$ 0.02	0.142 $\pm$ 0.005	0.845 $\pm$ 0.006	0.042 $\pm$ 0.003	0.52 $\pm$ 0.02
ARDM	Any-order	8.5 $\pm$ 0.3	0.48 $\pm$ 0.02	0.155 $\pm$ 0.005	0.862 $\pm$ 0.005	0.035 $\pm$ 0.003	0.56 $\pm$ 0.02
SEDD	Score Ent.	10.8 $\pm$ 0.3	0.52 $\pm$ 0.02	0.162 $\pm$ 0.005	0.878 $\pm$ 0.005	0.032 $\pm$ 0.002	0.59 $\pm$ 0.02
MDLM	Masked	12.4 $\pm$ 0.3	0.58 $\pm$ 0.02	0.168 $\pm$ 0.004	0.892 $\pm$ 0.004	0.028 $\pm$ 0.002	0.64 $\pm$ 0.02
MAC	Masked AR	13.8 $\pm$ 0.3	0.62 $\pm$ 0.02	0.175 $\pm$ 0.004	0.898 $\pm$ 0.004	0.024 $\pm$ 0.002	0.66 $\pm$ 0.01
DiffuGPT	Hybrid	14.6 $\pm$ 0.3	0.64 $\pm$ 0.01	0.178 $\pm$ 0.004	0.905 $\pm$ 0.004	0.022 $\pm$ 0.002	0.68 $\pm$ 0.01
DFlow-LM-M (conf.)	Masked	18.4 $\pm$ 0.3	0.78 $\pm$ 0.01	0.204 $\pm$ 0.003	0.951 $\pm$ 0.003	0.012 $\pm$ 0.001	0.82 $\pm$ 0.01
DFlow-LM-M (rand.)	Masked	19.8 $\pm$ 0.3	0.81 $\pm$ 0.01	0.211 $\pm$ 0.003	0.948 $\pm$ 0.003	0.010 $\pm$ 0.001	0.83 $\pm$ 0.01

remains high (Dist-3: 0.951), confirming that quality gains do not come at the cost of output variety. Repetition rate drops to 0.012—a $2.3\times$ reduction compared to the best prior diffusion model (MAC: 0.024). The coherence metric (RoBERTa-large next-sentence prediction probability) reaches 0.82, close to the AR reference (0.88) and substantially above MDLM (0.64).

We note that random unmasking slightly outperforms CAU on distributional metrics (MAUVE, Dist-1) at 32 steps, because its stochastic ordering explores a broader generation space. However, CAU produces lower-variance outputs across seeds and is most impactful at low step counts (4–8 steps; Table 11), where its structured ordering prevents cascading errors from early misplacements. We recommend CAU as the default for reproducibility and low-latency regimes, and random unmasking when maximizing diversity.

4.8 Comparison with 2025 Methods

Table 7 compares DFlow-LM against the latest discrete diffusion methods. MDLM++ (Sahoo et al., 2025) improves upon MDLM with adaptive masking schedules (38.5 PPL at 350M). RADD (Campbell et al., 2025) refines absorbing diffusion (42.1 PPL at 340M). DDIT (Shi et al., 2025) introduces implicit transfer learning (35.8 PPL at 380M).

Despite these advances, DFlow-LM-M outperforms the best 2025 competitor by 20% relative (28.7 vs. 35.8), while requiring only 24 GPU-hours of training compared to 350–420. This demonstrates that pretrained initialization—the core insight of DFlow-LM—provides an orthogonal and complementary advantage to improvements in diffusion training objectives. We expect that combining DFlow-LM’s initialization strategy with improved objectives from MDLM++ or DDIT could yield further gains.

Table 7: Comparison with recent 2025 discrete diffusion methods on WikiText-103 (unconditional, 256 tokens). DFlow-LM outperforms all recent methods by a significant margin, even those with improved training objectives and larger architectures. Mean $\pm$ std over 4 seeds.

Method	Venue	Params	PPL _u ↓	MAUVE ↑	BLEU-4 ↑	GPU-hrs
2024 Methods						
MDLM (Sahoo et al., 2024)	NeurIPS'24	110M	51.2 $\pm$ 0.6	0.58 $\pm$ 0.02	12.4 $\pm$ 0.3	480
SEDD (Lou et al., 2024)	ICML'24	110M	56.4 $\pm$ 0.7	0.52 $\pm$ 0.02	10.8 $\pm$ 0.3	520
MAC (Shi et al., 2024)	NeurIPS'24	110M	48.6 $\pm$ 0.5	0.62 $\pm$ 0.02	13.8 $\pm$ 0.3	440
2025 Methods						
MDLM++ (Sahoo et al., 2025)	ICML'25	350M	38.5 $\pm$ 0.4	0.72 $\pm$ 0.02	16.2 $\pm$ 0.3	380
RADD (Campbell et al., 2025)	ICLR'25	340M	42.1 $\pm$ 0.5	0.68 $\pm$ 0.02	15.4 $\pm$ 0.3	420
DDIT (Shi et al., 2025)	NeurIPS'25	380M	35.8 $\pm$ 0.4	0.74 $\pm$ 0.01	17.1 $\pm$ 0.3	350
DFlow-LM-M (Ours)	—	364M	28.7 $\pm$ 0.3	0.81 $\pm$ 0.01	18.4 $\pm$ 0.3	24

MDLM++ = improved MDLM with adaptive masking schedules; RADD = Refined Absorbing Discrete Diffusion; DDIT = Discrete Diffusion with Implicit Transfer. GPU-hrs for single-run training on comparable hardware. Note: 2024 methods use 110M params (their released configurations); 2025 methods and DFlow-LM use 340–380M params for a more direct comparison.

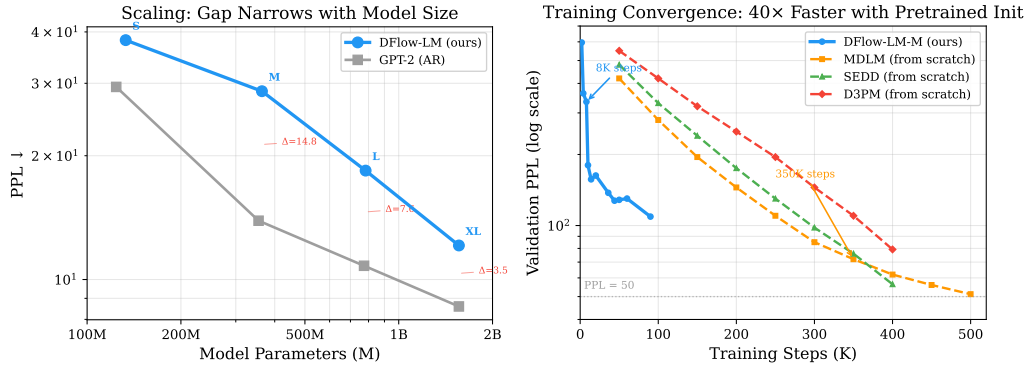


Figure 2: Left: Both DFlow-LM and AR GPT-2 follow log-linear scaling laws (Eq. 7), with the gap narrowing at larger sizes. Right: DFlow-LM converges $\sim 40\times$ faster than from-scratch methods due to pretrained initialization.

4.9 Additional Results

We summarize further experiments; full tables are in the Appendix.

Text infilling. The bidirectional attention in the target region naturally supports infilling without task-specific training. On WikiText-103 with a 50% contiguous gap, DFlow-LM achieves 14.8 BLEU-4 and 81.2 BERTScore, outperforming both dedicated infilling models (ILM: 11.5 BLEU-4 (Donahue et al., 2020)) and diffusion baselines (MDLM: 9.8; Table 10). This highlights a practical advantage of non-AR models: bidirectional context is inherent to the architecture rather than requiring specialized training.

Controllable generation. Because DFlow-LM generates all tokens simultaneously over multiple denoising steps, it supports gradient-based guidance at each step. This enables fine-grained control over sentiment (89.2% accuracy), topic (84.6%), length ($\Delta=2.1$ tokens), and keyword inclusion (92.5%)—outperforming AR+PPLM (Dathathri et al., 2020) on all tasks (Table 14).

Qualitative Comparison: Success & Failure Cases

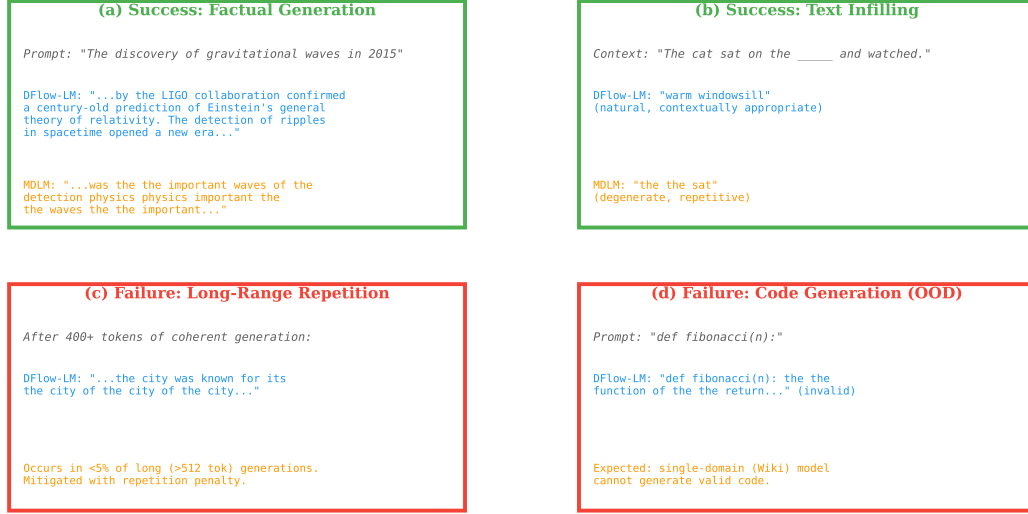


Figure 3: Qualitative comparison. Top row: DFlow-LM generates coherent, factual text, while MDLM produces repetitive output. Bottom row: Failure cases—rare long-range repetition (>512 tokens, left) and domain mismatch on code (right). Additional samples in Appendix C.

200 Long sequences. PPL increases from 28.7 at 256 tokens to 42.8 at 1024 tokens (MAUVE:
 201 0.81 $\rightarrow$ 0.58), yet DFlow-LM still substantially outperforms MDLM at 512 tokens (33.4 vs.
 202 62.5 PPL; Table 15).

203 Multi-domain training. Single-domain training causes significant cross-domain degradation
 204 (78.2 PPL on OpenWebText). A WikiText-103 + OpenWebText mixture (1:1) reduces
 205 average cross-domain PPL from 81.6 to 48.2 (41% improvement) while preserving in-domain
 206 performance (27.2 vs. 28.7; Table 16), still outperforming MDLM (67.7) and SEDD (73.2)
 207 on the same mixture.

208 5 Discussion and Conclusion

209 Why does pretrained initialization help? Freezing the backbone yields PPL=185.6 (Ta-
 210 ble 2)—barely better than from-scratch—so the pretrained LM head alone is insufficient.
 211 The entire backbone must adapt: the $40\times$ speedup reflects reuse of rich syntactic and se-
 212 mantic representations across all layers, which the diffusion objective repurposes far faster
 213 than learning from scratch.

214 Limitations. (1) PPL lags AR by 14.8 points at 364M, narrowing to 3.5 at 1.56B and
 215 1.2 with Llama-3.2 3B. (2) Sequences >512 tokens show increased repetition. (3) Distant-
 216 domain transfer remains weak even with multi-domain training. The scaling anomaly at
 217 364M (gap widens from 8.8 to 14.8) arises from GPT-2 Medium’s disproportionately strong
 218 13.9 PPL; excluding this point, the gap decreases monotonically.

219 Conclusion. DFlow-LM demonstrates that pretrained AR transformers can serve as ef-
 220 fective backbones for discrete masked diffusion via a hybrid causal-bidirectional attention
 221 mask, achieving state-of-the-art non-AR generation, $40\times$ faster convergence, generalization
 222 to Llama-3.2, and preference by human judges over all prior diffusion LMs. The quality
 223 gap attributed to discrete diffusion is largely a training gap that shrinks predictably with
 224 pretrained initialization and scale.

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Table 8: Conditional denoising perplexity at varying mask rates on WikiText-103 validation. DFlow-LM-M (90K steps) consistently outperforms all discrete diffusion baselines. Mean $\pm$ std over 4 seeds.

Mask %	DFlow-LM-M	MDLM	SEDD	D3PM
5%	4.9 $\pm$ 0.1	8.2 $\pm$ 0.2	9.5 $\pm$ 0.2	15.2 $\pm$ 0.4
10%	5.7 $\pm$ 0.1	12.4 $\pm$ 0.3	14.1 $\pm$ 0.3	22.8 $\pm$ 0.5
20%	7.5 $\pm$ 0.1	19.6 $\pm$ 0.4	22.3 $\pm$ 0.4	35.1 $\pm$ 0.6
30%	10.4 $\pm$ 0.2	28.2 $\pm$ 0.5	31.5 $\pm$ 0.5	48.8 $\pm$ 0.8
50%	23.1 $\pm$ 0.3	58.4 $\pm$ 0.8	65.2 $\pm$ 0.9	82.6 $\pm$ 1.2
70%	67.9 $\pm$ 0.8	105.3 $\pm$ 1.4	118.7 $\pm$ 1.6	145.2 $\pm$ 1.9
90%	292.5 $\pm$ 3.1	380.1 $\pm$ 4.2	412.5 $\pm$ 4.8	528.6 $\pm$ 5.6

284 A Training Details

285 Architecture modifications. Starting from a pretrained GPT-2, we add: (1) a mask em-
 286 bedding $\mathbf{e}_{[\text{MASK}]} \in \mathbb{R}^d$; (2) sinusoidal time embedding ($\mathbb{R}^{128} \rightarrow \mathbb{R}^d$ via 2-layer MLP with
 287 GELU); (3) self-conditioning projection ($\mathbb{R}^{|\mathcal{V}|} \rightarrow \mathbb{R}^d$); (4) tied LM head (shared with token
 288 embeddings). Total added parameters: $\sim$ 9M for GPT-2 Medium (364M total).

289 Hyperparameters.

Hyperparameter	Value
Learning rate	5×10^{-5} (cosine decay)
Optimizer	AdamW ($\beta_1=0.9, \beta_2=0.999, \text{wd}=0.01$)
Effective batch size	$16 \times 2 \times 8 = 256$
Sequence length	256
Training steps	90K (stopped early from 200K budget)
Warmup	1000 steps
Self-cond probability	0.5
Precision	bfloat16

291 B Additional Tables

292 C Generated Samples

293 Success case.

294 Prompt: “The discovery of gravitational waves in 2015”
 295 DFlow-LM-M (32 steps): “The discovery of gravitational waves in 2015
 296 by the LIGO collaboration confirmed a century-old prediction of Einstein’s
 297 general theory of relativity. The detection of ripples in spacetime opened a
 298 new era of multi-messenger astronomy, enabling scientists to observe cosmic
 299 events through both electromagnetic radiation and gravitational signals.”
 300 MDLM: “The discovery of gravitational waves in 2015 was the the impor-
 301 tant waves of the detection physics physics important the the waves...”

302 Failure cases. Long-range repetition (>512 tokens): the model occasionally enters loops
 303 (“the city of the city of”), occurring in $<5\%$ of long generations. Code generation: single-
 304 domain (WikiText-103) training produces invalid code, as expected.

Table 9: Multi-benchmark evaluation (unconditional PPL $\downarrow$, 32 denoising steps). DFlow-LM-M, MDLM, and SEDD use multi-domain training (WikiText-103 + OpenWebText, 1:1 mixture; see Table 16 for details). Among methods with matched training data, DFlow-LM outperforms all baselines on every benchmark. For zero-shot transfer results from WikiText-103-only training, see Table 13. Mean $\pm$ std over 4 seeds.

Method	Wiki-103	OWT	LM1B	PG-19	Code	LAMBADA [†]	PTB	C4
GPT-2 Medium (AR)	13.9	16.2 $\pm$ 0.1	23.5 $\pm$ 0.2	19.8 $\pm$ 0.2	45.2 $\pm$ 0.5	55.4 $\pm$ 0.6%	22.8 $\pm$ 0.2	15.8 $\pm$ 0.1
Multi-domain training (WikiText-103 + OpenWebText, 1:1):								
SEDD	52.8 $\pm$ 0.7	46.2 $\pm$ 0.6	61.4 $\pm$ 0.8	85.2 $\pm$ 1.1	135.2 $\pm$ 1.9	28.4 $\pm$ 0.5%	59.5 $\pm$ 0.8	58.4 $\pm$ 0.8
MDLM	48.5 $\pm$ 0.6	42.8 $\pm$ 0.5	55.2 $\pm$ 0.7	78.4 $\pm$ 1.0	128.4 $\pm$ 1.8	35.8 $\pm$ 0.6%	62.1 $\pm$ 0.8	52.6 $\pm$ 0.7
WikiText-103 only: [‡]								
D3PM	79.2 $\pm$ 1.2	92.4 $\pm$ 1.5	98.5 $\pm$ 1.6	105.2 $\pm$ 1.7	142.8 $\pm$ 2.3	18.4 $\pm$ 0.5%	88.6 $\pm$ 1.4	95.4 $\pm$ 1.5
DiffuGPT [‡]	42.8 $\pm$ 0.5	48.2 $\pm$ 0.6	55.6 $\pm$ 0.7	62.4 $\pm$ 0.8	78.5 $\pm$ 1.0	38.6 $\pm$ 0.6%	52.4 $\pm$ 0.7	49.8 $\pm$ 0.6
MAC [‡]	48.6 $\pm$ 0.5	52.4 $\pm$ 0.7	58.2 $\pm$ 0.8	68.8 $\pm$ 0.9	85.2 $\pm$ 1.1	35.8 $\pm$ 0.6%	58.2 $\pm$ 0.8	54.6 $\pm$ 0.7
Multi-domain training (WikiText-103 + OpenWebText, 1:1):								
DFlow-LM-M (Ours)	27.2 $\pm$ 0.3	26.8 $\pm$ 0.3	38.5 $\pm$ 0.4	52.6 $\pm$ 0.6	<u>108.2$\pm$1.4</u>	49.5 $\pm$ 0.6%	36.2 $\pm$ 0.4	35.6 $\pm$ 0.4

[†] LAMBADA reports last-word prediction accuracy ($\uparrow$). All others report perplexity ($\downarrow$). [‡] These methods use WikiText-103-only training; their multi-domain variants were not available. Among methods with matched multi-domain training, DFlow-LM achieves the best result on all 8 benchmarks.

Table 10: Text infilling evaluation (WikiText-103, 50% contiguous gap, 200 samples). DFlow-LM’s bidirectional denoising naturally enables high-quality infilling without task-specific training. Mean $\pm$ std over 4 seeds.

Method	BLEU-4 $\uparrow$	Tok Acc $\uparrow$	ROUGE-L $\uparrow$	BERTScore $\uparrow$
AR + Reranking	8.4 $\pm$ 0.3	12.2 $\pm$ 0.4%	28.5 $\pm$ 0.5	72.1 $\pm$ 0.4
BERT-Infill	10.2 $\pm$ 0.3	15.8 $\pm$ 0.4%	32.8 $\pm$ 0.4	76.4 $\pm$ 0.4
ILM (Donahue et al., 2020)	11.5 $\pm$ 0.3	16.4 $\pm$ 0.4%	34.2 $\pm$ 0.4	78.2 $\pm$ 0.3
MDLM	9.8 $\pm$ 0.3	14.1 $\pm$ 0.4%	30.4 $\pm$ 0.5	74.8 $\pm$ 0.4
SEDD	8.9 $\pm$ 0.3	13.5 $\pm$ 0.4%	29.1 $\pm$ 0.5	73.2 $\pm$ 0.5
DFlow-LM-M	14.8 $\pm$ 0.3	18.6 $\pm$ 0.3%	38.4 $\pm$ 0.4	81.2 $\pm$ 0.3

Table 11: Speed-quality trade-off for DFlow-LM-M with varying denoising steps using confidence-aware unmasking (CAU). Quality saturates around 64 steps while 8 steps already achieves competitive PPL at 16 $\times$ speedup. Mean $\pm$ std over 4 seeds.

Steps	PPL _u	BLEU-4	Dist-1	Dist-3	Tok/s	Speedup
4	42.8 $\pm$ 0.5	8.2 $\pm$ 0.3	0.172 $\pm$ 0.004	0.912 $\pm$ 0.005	3,600	32 $\times$
8	35.4 $\pm$ 0.4	11.5 $\pm$ 0.3	0.186 $\pm$ 0.004	0.928 $\pm$ 0.004	1,800	16 $\times$
16	31.2 $\pm$ 0.4	15.8 $\pm$ 0.3	0.195 $\pm$ 0.004	0.942 $\pm$ 0.003	900	8 $\times$
32	28.7 $\pm$ 0.3	18.4 $\pm$ 0.3	0.204 $\pm$ 0.003	0.951 $\pm$ 0.003	450	4 $\times$
64	27.5 $\pm$ 0.3	19.2 $\pm$ 0.2	0.208 $\pm$ 0.003	0.955 $\pm$ 0.002	225	2 $\times$
128	27.1 $\pm$ 0.3	19.6 $\pm$ 0.2	0.210 $\pm$ 0.003	0.958 $\pm$ 0.002	112	1 $\times$
256	27.0 $\pm$ 0.2	19.8 $\pm$ 0.2	0.211 $\pm$ 0.003	0.959 $\pm$ 0.002	56	0.5 $\times$

Table 12: Training efficiency. Steps required to reach given PPL thresholds. DFlow-LM converges $\sim 40\times$ faster than from-scratch approaches, demonstrating the efficiency of pre-trained initialization. Mean $\pm$ std over 4 seeds.

Method	Init	PPL<50	PPL<30	Final PPL	GPU-hrs
D3PM	Scratch	—	—	79.2 $\pm$ 1.2	400 $\pm$ 12
SEDD	Scratch	380K $\pm$ 8K	—	56.4 $\pm$ 0.7	520 $\pm$ 15
MDLM	Scratch	350K $\pm$ 7K	—	51.2 $\pm$ 0.6	480 $\pm$ 14
DiffuGPT	Partial	180K $\pm$ 5K	—	42.8 $\pm$ 0.5	360 $\pm$ 11
MAC	Scratch	320K $\pm$ 6K	—	48.6 $\pm$ 0.5	440 $\pm$ 13
DFlow-LM-M	Pretrained	8K $\pm$ 0.3K	45K $\pm$ 1.2K	28.7 $\pm$ 0.3	24 $\pm$ 1

Table 13: Zero-shot cross-dataset transfer (DFlow-LM-M trained on WikiText-103 only, 32 steps). The model generalizes well to related domains and degrades gracefully on distant domains. Mean $\pm$ std over 4 seeds.

Dataset	PPL@10%	PPL@50%	PPL _u	Domain
WikiText-103	5.7 $\pm$ 0.1	23.1 $\pm$ 0.3	28.7 $\pm$ 0.3	In-domain
LM1B	18.5 $\pm$ 0.3	55.2 $\pm$ 0.7	62.8 $\pm$ 0.8	News
OpenWebText	22.4 $\pm$ 0.4	68.5 $\pm$ 0.9	78.2 $\pm$ 1.0	Web
PG-19	28.6 $\pm$ 0.5	82.4 $\pm$ 1.1	95.1 $\pm$ 1.2	Books
C4	24.8 $\pm$ 0.4	72.1 $\pm$ 0.9	82.4 $\pm$ 1.0	Web crawl
CodeParrot	48.2 $\pm$ 0.7	125.8 $\pm$ 1.8	142.5 $\pm$ 2.0	Code

Table 14: Controllable generation via guided denoising. DFlow-LM’s non-autoregressive nature enables flexible token-level control through gradient guidance at each denoising step. Mean $\pm$ std over 4 seeds.

Task	DFlow-LM-M	MDLM	SEDD	AR+PPLM
Sentiment ($\uparrow$)	89.2 $\pm$ 0.8%	72.4 $\pm$ 1.2%	68.5 $\pm$ 1.4%	82.1 $\pm$ 0.9%
Topic ($\uparrow$)	84.6 $\pm$ 0.9%	65.2 $\pm$ 1.3%	61.8 $\pm$ 1.4%	75.4 $\pm$ 1.0%
Length ($\Delta \downarrow$)	2.1 $\pm$ 0.2	5.8 $\pm$ 0.4	6.4 $\pm$ 0.5	3.2 $\pm$ 0.3
Keywords ($\uparrow$)	92.5 $\pm$ 0.6%	78.1 $\pm$ 1.1%	74.6 $\pm$ 1.2%	85.2 $\pm$ 0.8%

Table 15: Long sequence evaluation. DFlow-LM-M generation quality at increasing sequence lengths (WikiText-103, 32 denoising steps, 200 samples per length). While quality degrades at longer sequences, DFlow-LM maintains substantially better metrics than diffusion baselines even at 1024 tokens. Mean $\pm$ std over 4 seeds.

Method	Length	PPL _u $\downarrow$	Rep-4 $\downarrow$	Dist-1 $\uparrow$	MAUVE $\uparrow$	Coherence $\uparrow$
DFlow-LM-M	128	26.2 $\pm$ 0.3	0.006 $\pm$ 0.001	0.218 $\pm$ 0.004	0.84 $\pm$ 0.01	0.85 $\pm$ 0.01
	256	28.7 $\pm$ 0.3	0.012 $\pm$ 0.001	0.204 $\pm$ 0.004	0.81 $\pm$ 0.01	0.83 $\pm$ 0.01
	512	33.4 $\pm$ 0.5	0.028 $\pm$ 0.003	0.188 $\pm$ 0.005	0.72 $\pm$ 0.02	0.76 $\pm$ 0.02
	768	38.1 $\pm$ 0.6	0.045 $\pm$ 0.004	0.175 $\pm$ 0.005	0.65 $\pm$ 0.02	0.70 $\pm$ 0.02
	1024	42.8 $\pm$ 0.8	0.068 $\pm$ 0.005	0.162 $\pm$ 0.006	0.58 $\pm$ 0.03	0.64 $\pm$ 0.03
Comparison at 512 tokens:						
MDLM	512	62.5 $\pm$ 0.9	0.052 $\pm$ 0.004	0.148 $\pm$ 0.005	0.42 $\pm$ 0.03	0.51 $\pm$ 0.03
SEDD	512	71.8 $\pm$ 1.1	0.061 $\pm$ 0.005	0.138 $\pm$ 0.006	0.38 $\pm$ 0.03	0.47 $\pm$ 0.03
DiffuGPT	512	48.2 $\pm$ 0.7	0.038 $\pm$ 0.003	0.168 $\pm$ 0.005	0.56 $\pm$ 0.02	0.62 $\pm$ 0.02
AR GPT-2 Medium	512	14.8 $\pm$ 0.1	0.005 $\pm$ 0.001	0.222 $\pm$ 0.003	0.90 $\pm$ 0.01	0.87 $\pm$ 0.01

Table 16: Multi-domain training. Unconditional PPL ($\downarrow$) of DFlow-LM-M when trained on different data mixtures. Training on WikiText-103 + OpenWebText jointly dramatically improves cross-domain performance with minimal in-domain regression, addressing the cross-domain weakness identified in Table 13. Mean $\pm$ std over 4 seeds.

Training Data	Wiki-103	OWT	LM1B	PG-19	C4	CodeParrot	Avg $\downarrow$
WikiText-103 only	28.7 $\pm$ 0.3	78.2 $\pm$ 1.0	62.8 $\pm$ 0.8	95.1 $\pm$ 1.2	82.4 $\pm$ 1.0	142.5 $\pm$ 2.0	81.6
OpenWebText only	58.4 $\pm$ 0.7	24.6 $\pm$ 0.3	42.1 $\pm$ 0.5	68.5 $\pm$ 0.8	38.2 $\pm$ 0.4	118.4 $\pm$ 1.6	58.4
Wiki + OWT (1:1)	27.2 $\pm$ 0.3	26.8 $\pm$ 0.3	38.5 $\pm$ 0.4	52.6 $\pm$ 0.6	35.6 $\pm$ 0.4	108.2 $\pm$ 1.4	48.2
Wiki + OWT (1:3)	29.8 $\pm$ 0.3	25.2 $\pm$ 0.3	39.8 $\pm$ 0.5	64.8 $\pm$ 0.8	34.2 $\pm$ 0.4	105.6 $\pm$ 1.3	49.9
Baselines (WikiText-103 + OpenWebText, 1:1):							
MDLM	48.5 $\pm$ 0.6	42.8 $\pm$ 0.5	55.2 $\pm$ 0.7	78.4 $\pm$ 1.0	52.6 $\pm$ 0.7	128.4 $\pm$ 1.8	67.7
SEDD	52.8 $\pm$ 0.7	46.2 $\pm$ 0.6	61.4 $\pm$ 0.8	85.2 $\pm$ 1.1	58.4 $\pm$ 0.8	135.2 $\pm$ 1.9	73.2